

HealthConnect Clinic

Data Analytics Track — Week 4 Initial Analysis Document

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1. Understanding the Project Brief

This document responds to the AnalystLab Africa Experience Lab Week 4 assignment: "HealthConnect Project Kickoff & Problem Understanding." HealthConnect Clinic is a fictional outpatient healthcare provider facing missed appointments, unclear no-show drivers, inefficient use of appointment slots, repetitive patient enquiries, and a need to improve patient engagement and administrative support. The central project question set by the brief is:

"How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?"

This is Week 4 of the internship programme, not Week 1 of the project — Weeks 1–3 covered general track skill-building and are not repeated here. HealthConnect work begins fresh at Week 4, and this document is that Week 4 output for the Data Analytics track.

- **Track-specific problem:** As the Data Analytics track, our responsibility is to understand the appointment data and identify how it can be used to investigate appointment attendance and no-show patterns — laying the analytical foundation the wider project (including the Data Science and Generative AI tracks) will build on.
- **Deliverable expected of this track:** An Initial Analysis Document/Notebook: dataset overview, data-quality assessment, business questions, 3–5 justified KPIs (identification only, not yet calculated per the brief), and an initial analysis approach — plus embedded assumptions, limitations, risks and dependencies.

Note: the "HealthConnect Clinic Knowledge Base" is also a supplied project resource, but its primary users are the Generative AI track (Healthcare Information Assistant design). It contains no appointment-level data and is not used in this Data Analytics output; it is retained in the project folder for cross-track reference only, unaltered.

2. Our Role as Data Analyst

For this engagement we act as the clinic's Data Analyst, embedded in an operations/administration function. The role is scoped as follows:

- **In scope:** Descriptive and diagnostic analysis of historical appointment data; data quality assurance; KPI definition; producing evidence-based recommendations for reducing no-shows.
- **Out of scope:** Clinical decision-making, patient diagnosis, or any interpretation of medical information — consistent with the safety boundaries defined for the clinic's other AI/analytics tools.
- **Working style:** Iterative and documented: each step of the analysis is recorded so that findings are reproducible and can be reviewed by a non-technical stakeholder (e.g. clinic operations manager).

3. Data Sources

Two source files were provided and used in this track:

File	Type	Contents	Size
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HealthConnect_Appointment_Data.csv	CSV	5,000 appointment records, 18 fields	~577 KB
HealthConnect_Data_Dictionary.xlsx	XLSX	Field-level definitions, types, examples, notes for all 18 fields	~6 KB

4. Project Folder Structure

A working folder was created to keep raw data, analysis notebooks, documentation and outputs separated:

```
HealthConnect_Project/  data/          raw CSV + Data Dictionary (unmodified)
notebooks/            Week 4 exploratory analysis notebook (.ipynb)  docs/          this
analysis document, Week 4 summary, README  outputs/      exported figures / summary
tables (as needed)
```

5. Data Dictionary Review

The Data Dictionary defines 18 fields. Reviewing it first (before touching the data) established expected types, valid value sets, and known caveats:

Field	Type	Key notes
appointment_id	Text	Primary key, unique per row
patient_id	Text	Anonymised; repeats across appointments
gender	Text	Synthetic/anonymised category
age / age_group	Int / Text	Adults only; age_group is a derived band
appointment_type	Text	4 categories
booking_date / appointment_date	Date	ISO format
appointment_day / appointment_time	Text	Derived weekday; time-of-day band
booking_lead_days	Int	Days between booking and appointment (derivable)
previous_appointments / previous_no_shows	Int	Historical counts; no-shows cannot exceed appointments
reminder_sent / reminder_channel	Text	Channel is blank when no reminder sent
distance_to_clinic_km / waiting_time_minutes	Decimal / Int	Both noted as having limited missing values
appointment_outcome	Text	Target variable: Attended / No-Show / Cancelled

6. Dataset Inspection

Initial inspection (full detail and code in the accompanying notebook) confirmed the data matches the dictionary:

- 5,000 rows × 18 columns; all columns present and correctly typed after parsing dates.
- 1,696 unique patients across 5,000 appointments — patients appear multiple times, so patient-level and appointment-level analysis will need to be distinguished.
- appointment_date ranges 1 Jan 2025 – 30 Jun 2026; booking_date ranges 7 Nov 2024 – 27 Jun 2026.
- No fully duplicated rows and no duplicate appointment_id values.

7. Initial Data-Quality Assessment

7.1 Completeness

Field	Missing	% Missing	Assessment
reminder_channel	1,366	27.3%	Expected — exactly matches rows where reminder_sent = "No". Not a data error.
distance_to_clinic_km	90	1.8%	Genuine gap; dictionary flags this as a known limitation.
waiting_time_minutes	60	1.2%	Genuine gap; dictionary flags this as a known limitation.
All other fields	0	0.0%	Complete.

7.2 Validity & Consistency

- previous_no_shows never exceeds previous_appointments (0 violations) — respects the dictionary's stated business rule.
- booking_lead_days recomputed from (appointment_date – booking_date) matches the provided column in all 5,000 rows; no negative lead times.
- appointment_day recomputed from appointment_date matches the provided column in all 5,000 rows.
- age_group recomputed from age matches the provided column in all 5,000 rows.
- No negative values in distance_to_clinic_km or waiting_time_minutes.

7.3 Anomalies for Follow-Up

- 737 appointments fall on a Sunday. This is not a data error within the file itself (dates and day-of-week are internally consistent) — it should simply be noted as an open question about clinic operating-hours assumptions, since it does not affect analysis of the outcome variable but should be flagged if this dataset is ever reconciled against real clinic opening hours.
- gender includes a small "Prefer not to say" category (108 rows, 2.2%) — retain as its own category rather than imputing.

Overall verdict: data quality is high. No records need to be dropped; missingness is limited, largely explainable, and concentrated in two non-critical numeric fields that can be handled with documented imputation or exclusion in later analysis.

8. Important Variables Identified

An early association scan (outcome rates split by each field — see notebook) was used to prioritise which variables matter most for explaining no-shows:

Variable	Signal strength	Observation
booking_lead_days	Strong	No-show rate rises from 28% (0–7 days lead) to 68% (46–60 days lead) — the clearest single predictor found.
previous_no_shows	Strong	No-show rate rises from 44% (0 prior no-shows) to 68–100% (3+ prior no-shows).
reminder_sent	Moderate	No-show rate is ~4 points lower when a reminder was sent (47.4% vs 51.4%).
reminder_channel	Weak–moderate	SMS shows a slightly lower no-show rate (45.8%) than WhatsApp (49.8%) or Email (48.4%).
appointment_type / age_group / appointment_time / gender	Weak	No-show rates are broadly similar (within a few points) across categories — unlikely to be primary drivers on their own.
distance_to_clinic_km / waiting_time_minutes	Weak (so far)	Means are similar across outcome groups; may still interact with other variables — worth deeper testing.
appointment_outcome	Target	48.5% No-Show / 46.3% Attended / 5.3% Cancelled — this is the variable we are ultimately trying to explain.

9. Business Questions

- Q1. What is the overall no-show rate, and how has it trended by month/quarter across the data period?
- Q2. Which factors are most strongly associated with a patient missing their appointment — specifically booking lead time, reminder usage/channel, and prior no-show history?
- Q3. Are certain appointment types, time slots, or days more prone to no-shows, and could scheduling policy reduce this?
- Q4. Does sending a reminder — and the channel used — measurably change the likelihood of attendance?
- Q5. Which patient segments (by prior no-show count, distance, or age group) should be prioritised for a targeted intervention (e.g. earlier reminders, confirmation calls)?

10. Key Performance Indicators (KPIs)

Per the Week 4 brief, KPIs at this stage are identified and justified only — they are not yet calculated, analysed, or visualised. Each KPI below is explicitly linked to the business question(s) it will help answer in later stages.

KPI	Definition	Linked question(s)	Why it's relevant
No-Show Rate	No-Show appointments ÷ total scheduled appointments	Q1	Primary metric for the core business problem — the clinic's central pain point.

Attendance Rate	Attended ÷ total scheduled appointments	Q1	Complements no-show rate for capacity-planning context.
Cancellation Rate	Cancelled ÷ total scheduled appointments	Q1, Q3	Separates planned non-attendance from unplanned no-shows, which likely need different operational responses.
Reminder Effectiveness	No-show rate with reminder sent vs. not, split by channel	Q4	Directly measures whether the reminder programme — and which channel — changes attendance.
Repeat No-Show Rate	Share of appointments from patients with ≥2 prior no-shows	Q2, Q5	Identifies a high-risk segment worth prioritising for a targeted intervention.

11. Initial Analysis Approach

- Descriptive stage (Week 4, this notebook): outcome distribution, univariate breakdowns by each candidate variable, missingness and consistency checks — establishes the baseline and shortlists variables worth deeper testing.
- Diagnostic stage (Week 2): statistical association tests (chi-square for categorical variables, point-biserial/logistic coefficients for numeric variables) against appointment_outcome; segment cross-tabs (e.g. lead-time × prior no-shows).
- Predictive stage (optional, later): a simple classification model (e.g. logistic regression) to estimate no-show probability per appointment, using the strong/moderate-signal variables identified in Section 8 as candidate features.
- Reporting stage: translate findings into an operations-facing summary — KPI dashboard view plus 2–3 concrete, low-cost intervention recommendations (e.g. reminder timing, overbooking policy for high-risk slots).

12. Assumptions, Limitations & Risks

Assumptions

- The dataset is synthetic/fictional (per the accompanying Knowledge Base) — patterns are illustrative for the exercise and are not claims about a real clinic.
- "Appointment-level" analysis is the primary unit; patient-level rollups will be treated as a secondary lens because patients repeat in the data.
- Cancelled appointments are treated as a distinct outcome from No-Show throughout, since they represent different operational events (advance notice vs. none).

Limitations

- distance_to_clinic_km and waiting_time_minutes have small amounts of missing data (1.8% and 1.2%) that will need an explicit, documented handling rule before any modelling step.
- No clinician- or staff-level fields are present, so operational causes on the clinic side (e.g. understaffing on a given day) cannot be analysed from this data alone.

- The dataset covers appointment records only — it cannot explain patient intent or reasons for missing an appointment; findings are associative, not causal, unless tested more rigorously in later stages.

Risks

- Overfitting simple explanations to a synthetic dataset: findings should be framed as directional and hypothesis-generating for Week 4, not final conclusions.
- Two related fields could be mixed up in downstream work: `booking_lead_days` is provided directly and does not need to be recalculated — using both the raw dates and the provided column risks double-counting or inconsistent logic if not documented centrally.
- Sunday appointment records (Section 7.3) should be clarified with a stakeholder before any operating-hours-based segmentation is built, to avoid a false assumption entering later analysis.

Next: `notebooks/HealthConnect_Week4_EDA.ipynb` for the code, tables and charts referenced above, and `docs/HealthConnect_Week4_Project_Summary.docx` for the concise Part 4 submission summary. See `README.md` for full project navigation.